

Yucheng (Cecilia) Zhang

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EDUCATION

Northwestern University

M.S. in Computer Science

Aug. 2026 – May 2028

Evanston, IL

The University of Texas at Austin

B.S. in Mathematics, Minor in Computer Science | GPA: 3.74/4.00

Aug. 2023 – May 2026

Austin, TX

TECHNICAL SKILLS

ML/AI: Scikit-learn, PyTorch, TensorFlow, LLM APIs, LangChain, CrewAI, prompt engineering, model evaluation, AI feedback systems

Data Science: Pandas, NumPy, R, MATLAB, MNE, ggplot2, statistics, feature engineering, backtesting, ROC/AUC

Data/Cloud: SQL Server, PostgreSQL, Azure Synapse, Databricks, Delta Lake, AWS, Docker, Power BI, Linux

Domains: Credit risk, fraud detection, bioinformatics, EEG/fNIRS, computer vision, financial analysis

MACHINE LEARNING, AI, AND DATA EXPERIENCE

CodeAssist – AI Feedback System | GitHub

Aug. 2025 – Present

University of Texas at Austin | Flask, React, PostgreSQL, OpenAI, Gemini, Claude

Austin, TX

- Built assignment-level AI feedback controls, enabling instructors to configure prompt templates, approved context inputs, model settings, request limits, and cooldown rules.
- Implemented LLM context-filtering logic so only instructor-approved assignment descriptions, student code, test results, test cases, and student output are sent to AI providers.
- Improved multi-provider AI workflows across OpenAI, Gemini, Claude, and local models, including model selection, API configuration, timeout handling, and settings consistency fixes.

Enterprise Risk Analyst Intern

May 2025 – Dec. 2025

Toyota Industries Commercial Finance | Python, SQL, Azure, Power BI

Dallas, TX

- Built an end-to-end credit-risk modeling pipeline from SQL Server data, including ETL, feature engineering, Logistic Regression, and Random Forest training.
- Engineered features including PayNet inquiries, loan-to-value ratios, account tenure, and asset-segment signals for trucks, trailers, and used-vehicle loans.
- Validated models with ROC/AUC, confusion matrices, and feature importance, achieving AUC > 0.82 and improving risk-flagging accuracy by 3–4%.
- Backtested predicted risk buckets against observed outcomes, identifying up to 3.7× higher default rates in high-risk groups and informing scorecard refinements.

Researcher & TA – Biological Data Analytics

Jan. 2024 – Dec. 2024

University of Texas at Austin | R, Linux, AWS, Docker

Austin, TX

- Processed 250GB of RNA-seq data with Salmon, RSEM, Trinity, and samtools, achieving a 98.7% mapping rate across reproducible pipelines.
- Performed Voom/Limma differential-expression analysis, identifying 48 key genes across time-series samples with permutation-based FDR.
- Automated normalization, PCA, and volcano-plot workflows in R and ggplot2, improving interpretation efficiency by 15%.

FinChat – AI-Powered Financial Analysis System

Jan. 2023 – May 2024

Southern Methodist University | Python, FinGPT, LangChain, CrewAI

Remote

- Built LLM workflows for financial report analysis, extracting and processing 100+ SEC filings from EDGAR into a scalable data lake.
- Integrated FinGPT, LangChain, and CrewAI components for metric extraction, trend detection, and contextual financial interpretation, improving processing accuracy by 3%.

Data Scientist Assistant Intern – Image Restoration

Jul. 2023 – Sep. 2023

Chinese Academy of Sciences | TensorFlow, Cloud, GFPGAN

Beijing, China

- Deployed image restoration modules on Alibaba Cloud via SSH, implementing VAE loss, Gaussian filters, and LSGAN functions for scalable reconstruction workflows.
- Constructed a TensorFlow 2 CNN, split 3.4k images into train/validation/test sets, and achieved 80% accuracy across 5 image categories.

RESEARCH PROJECTS AND PUBLICATIONS

Neuroengineering Lab – Machine Learning Research Assistant

Jan. 2023 – May 2024

Python, MATLAB, MNE, RF, SVM, KNN, XGBoost

Remote

- Processed 35GB of EEG/fNIRS data for signal cleaning, feature extraction, connectivity analysis, and statistical testing; contributed to Frontiers and IEEE publications.
- Evaluated emotion-recognition models using RF, SVM, KNN, and XGBoost, strengthening experimental design, biomarker analysis, and applied ML research writing.

Kiwifruit Quality Grading – Machine Learning Member | Paper

Aug. 2022 – May 2024

Computer Vision, Feature Fusion, Model Evaluation

Published 2025

- Contributed to image-based grading with color, texture, shape, and fusion features; benchmarked models for a paper published in *Expert Systems with Applications*.
- Compared feature combinations to optimize grading accuracy and interpret model behavior for the paper's ML methodology section.

SELECTED PUBLICATIONS

- Expert Systems with Applications* (2025): kiwifruit quality grading using image multi-feature fusion and ResTNet.
- Frontiers in Neurology* (2024) and *Frontiers in Neuroscience* (2023): EEG/fNIRS cognitive impairment and digital therapy studies.